Frequentist vs Bayesian surgery-risk models

Same accuracy, same list. The difference is how each says how sure it is.

The bottom line

0.73

AUC for both on the 2025-07 test (0.733 vs 0.732)

98.1%

of the top-2,500 contact list is the same patients under both models

2.2%

predicted vs 2.2% observed in 2025-07, after adding time since first visit (was +30%)

What differs is uncertainty, not accuracy

- Both models give an interval around every prediction: a bootstrap confidence interval for the frequentist model, a posterior credible interval for the Bayesian one.
- The Bayesian model also answers "how sure are we this patient belongs on the list?": 1,799 patients are near-certain, 1,475 are a coin flip.
- Neither interval is better calibrated: in a simulation with known true risk, 89% intervals covered 89.5% (Bayesian) vs 88.0% (bootstrap).
- Recommendation: score with the frequentist model and use the Bayesian one to report uncertainty.

Two methods, identical inputs

Frequentist

- scikit-learn L2 logistic regression (C = 1)
- One best-fit set of coefficients
- Intervals from 200 patient-level bootstrap refits
- Fits in 0.04 s

Bayesian

- PyMC logistic regression, nutpie NUTS sampler
- N(0, 0.25) priors chosen by a prior predictive check
- 4,000 posterior draws of every coefficient
- Fits in 46 s, 0 divergences, r-hat ≤ 1.003

Shared: 36 standardized features (recent activity, BMI, age, conditions, and time since first back pain visit capped at 12m+), trained on 60,594 snapshots (2024-07 to 2025-04, 1,878 surgeries), tested out-of-time on 19,986 patients in 2025-07. Target: surgery while still covered within 12 months.

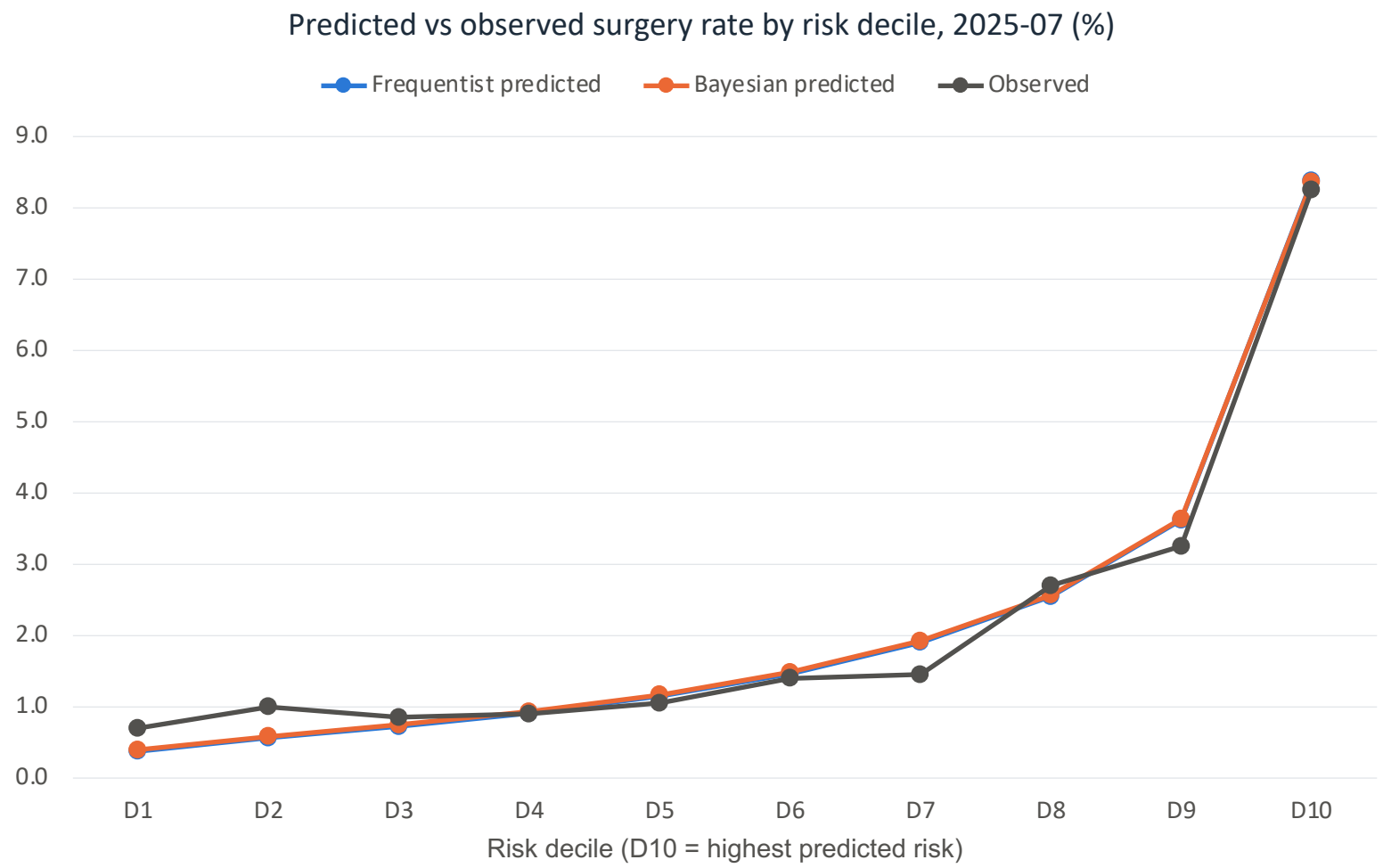
Accuracy

Out-of-time test on 2025-07: discrimination, calibration, and how many surgeries the list catches

Same accuracy on every metric

Metric	Frequentist	Bayesian	What it means
AUC	0.733	0.732	Chance a surgery patient is ranked above a non-surgery patient
Brier score	0.0203	0.0203	Mean squared error of the probabilities (lower is better)
Log loss	0.0944	0.0944	Penalizes confident wrong predictions (lower is better)
Calibration slope	0.96	0.97	1.00 = spread of risk is right; slightly below 1 is mild over-confidence
Mean predicted / observed	2.2% / 2.2%	2.2% / 2.2%	Calibrated in both (was 30% too high without the tenure feature)
Surgeries caught in top 2,500	41.8%	41.5%	Top 2,500 is 13% of patients, about Meridian's quarterly capacity
Fit time	0.04 s	46 s	Matters for automated retraining

Both models are calibrated out-of-time

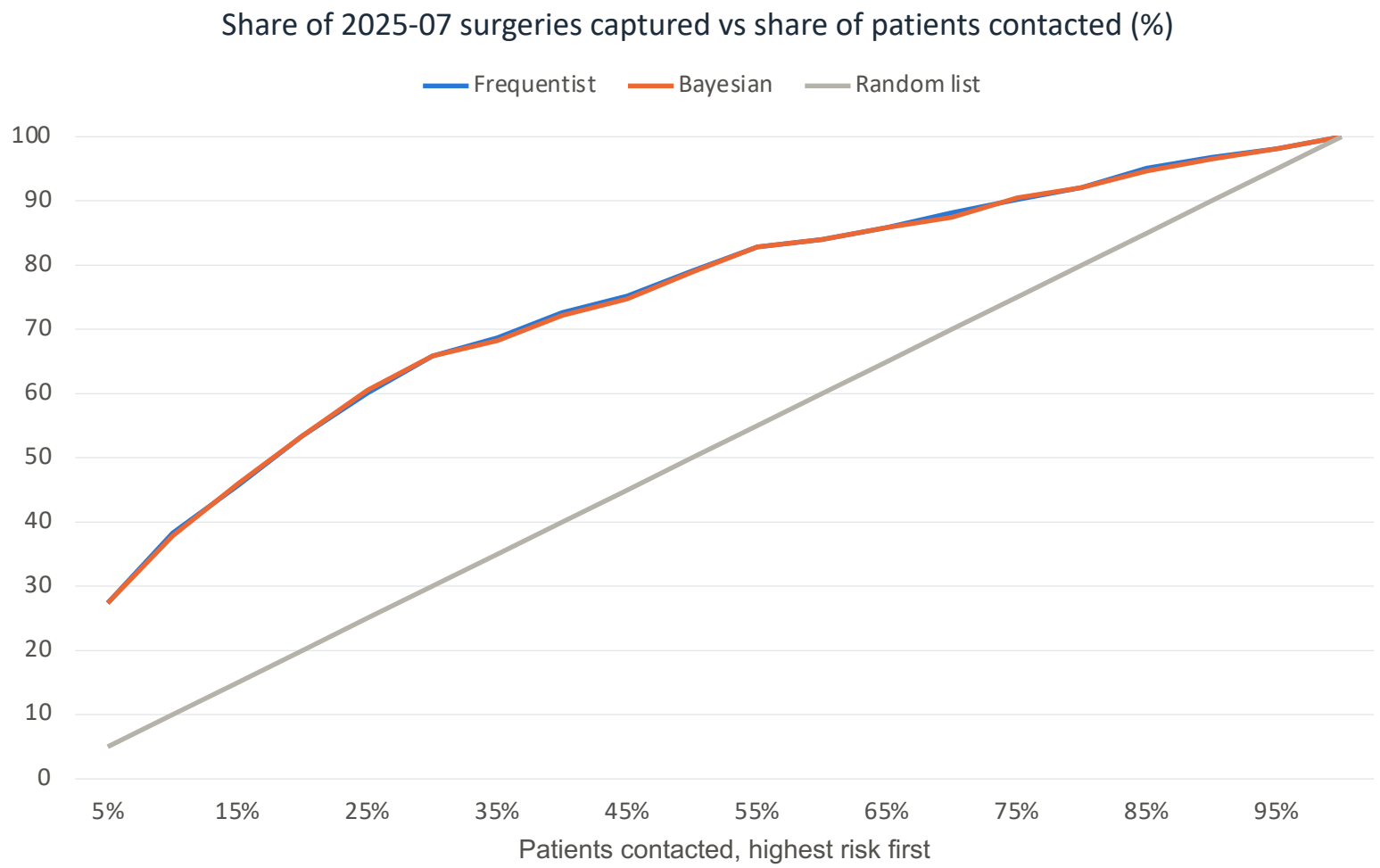


The two predicted lines overlap
The models agree decile by decile.

Both track observed
Overall 2.2% predicted vs 2.2% observed, even though risk fell from 3.1% in training to 2.2%.

Slight misses at the bottom
The lowest deciles run a little below observed (0.4% vs 0.7%). That doesn't matter for a top-of-list decision.

A 2,500-patient list catches about 42% of surgeries



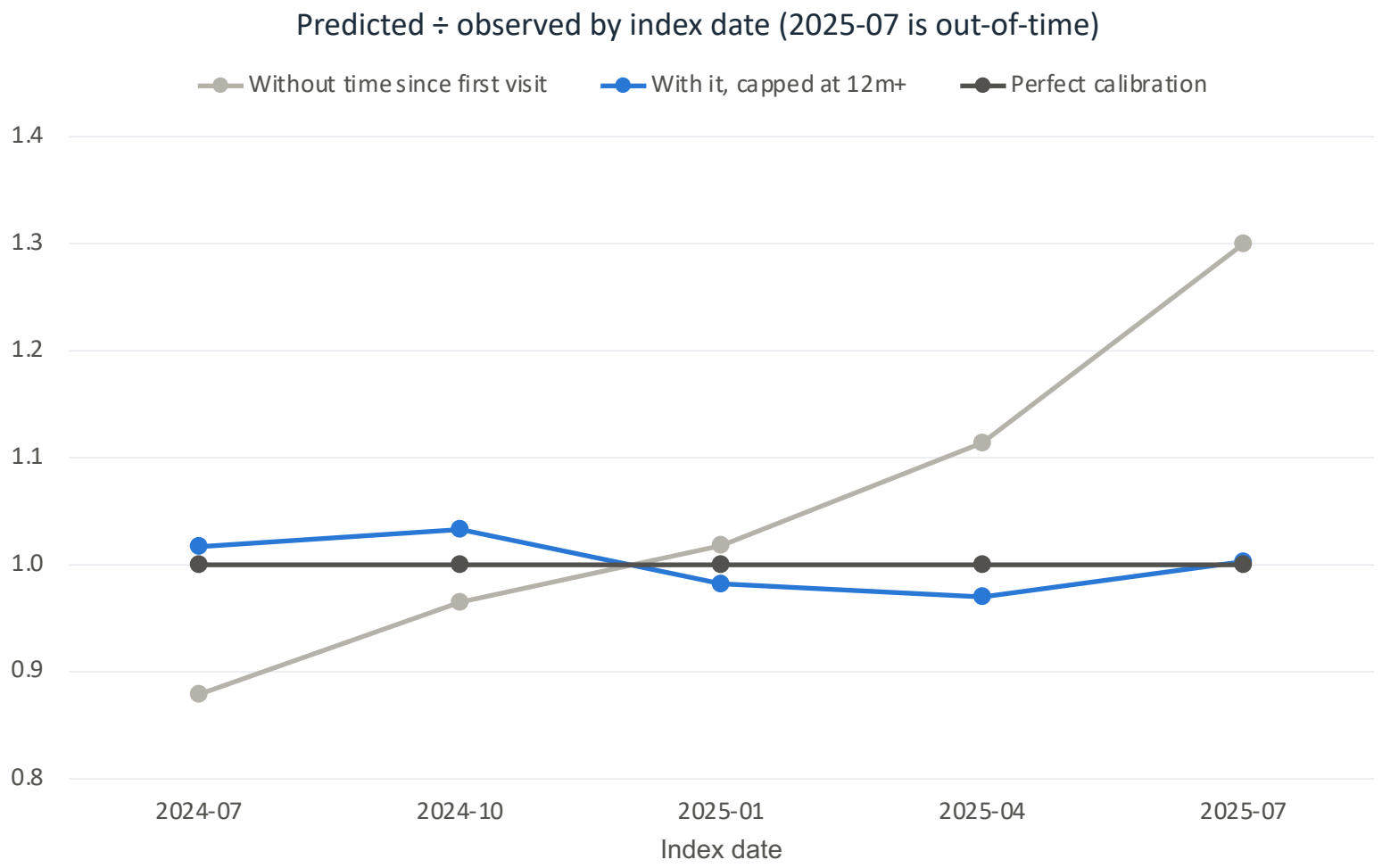
42%

of surgeries land in the top 2,500, for either model

3.3×

a random list's catch. The two models differ by -1 surgeries

The 30% over-prediction was a missing feature



What went wrong
Patients who reach a year without surgery are far less likely to have one. Without that feature, the model over-predicted them 2.5×.

Why it grew over time
The cohort is closed, so it ages: patients 12+ months out were 19% of training rows and 48% of the 2025-07 test.

Why capped
Every 2026-07 patient is 12m+, a band training covers well.

Five real patients

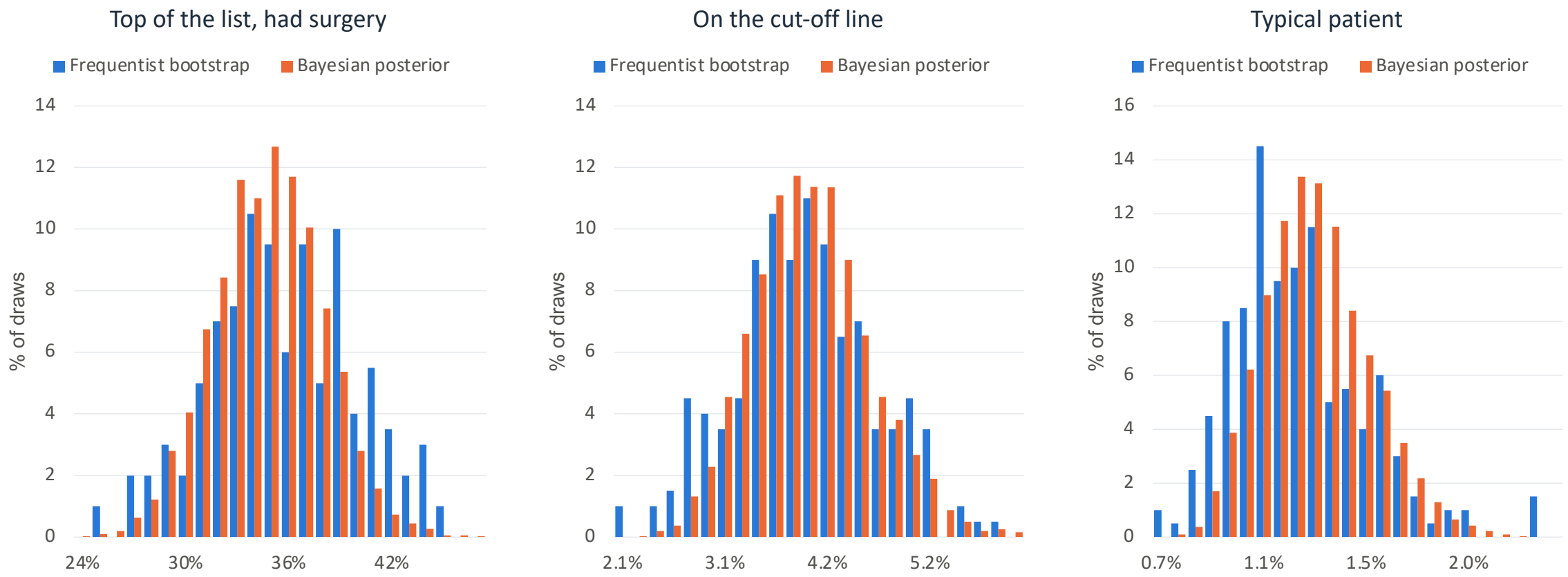
Point estimates, intervals, and the full distribution behind each prediction (2025-07 test period)

Point estimates agree; intervals differ

Situation	Profile	Outcome	Frequentist (89% CI)	Bayesian (89% CrI)	P(on list)
Top of the list, had surgery	57M, BMI not recorded, prior consult, recent imaging, 3 LBP visits/yr	**Surgery**	**35.3%** 28.7% – 42.5%	**34.2%** 29.4% – 39.3%	**1.00**
High risk, no surgery	50M, BMI not recorded, prior consult, recent imaging, 3 LBP visits/yr	No surgery	**24.0%** 18.4% – 29.8%	**24.1%** 20.1% – 28.1%	**1.00**
On the cut-off line	63M, BMI 35+, 3 LBP visits/yr	No surgery	**4.0%** 2.8% – 5.1%	**4.0%** 3.1% – 5.0%	**0.47**
Typical patient	64F, BMI not recorded, injection, 3 LBP visits/yr	No surgery	**1.3%** 0.9% – 1.7%	**1.3%** 1.0% – 1.7%	**0.00**
Missed: low risk, had surgery	77M, BMI not recorded, 5 LBP visits/yr	**Surgery**	**0.3%** 0.2% – 0.4%	**0.3%** 0.2% – 0.4%	**0.00**

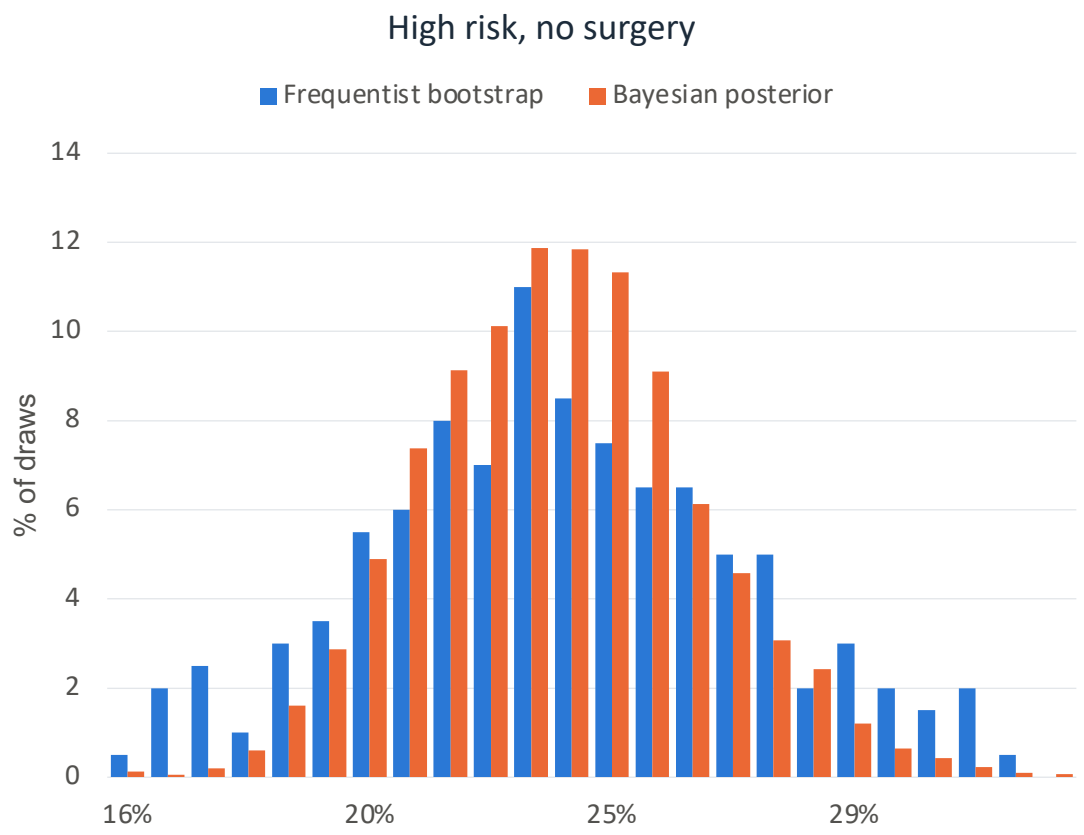
P(on list) is Bayesian only: the share of posterior draws in which the patient ranks in the top 2,500. The cut-off sits at 4.0% predicted risk. The frequentist interval comes from 200 bootstrap refits.

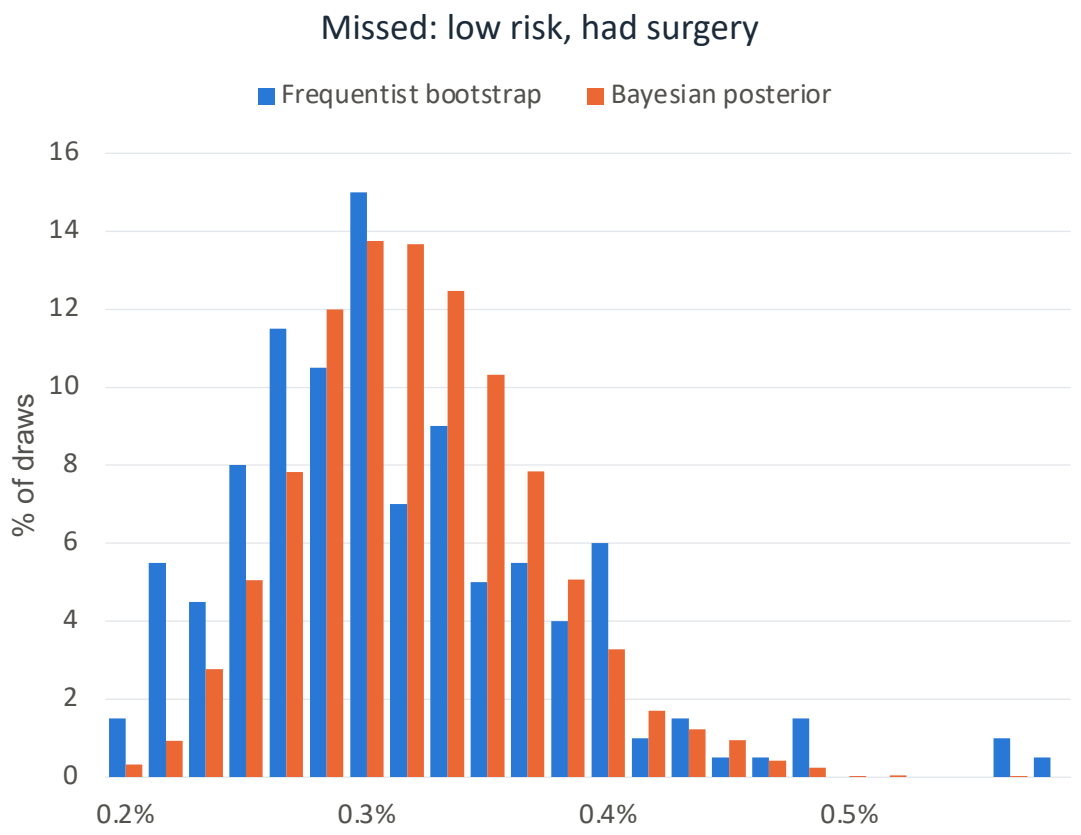
Each prediction is a distribution, not a point



Distribution of each patient's 12-month risk across 200 bootstrap refits (blue) and 4,000 posterior draws (orange). The x-axis is the predicted risk.

Same story for the patients the list gets wrong





A confident 24% risk still means most such patients don't have surgery. A tight interval around 0.3% still allows the surgery we missed. Intervals describe the risk, not the outcome.

Yes, the frequentist model has intervals too

Frequentist confidence interval

- Built from the coefficients' sampling uncertainty, either by bootstrap (used here) or by the delta method on the standard errors.
- "If we refit on many datasets like this one, 89% of intervals would contain the true risk."
- Median width here: 0.8%

Bayesian credible interval

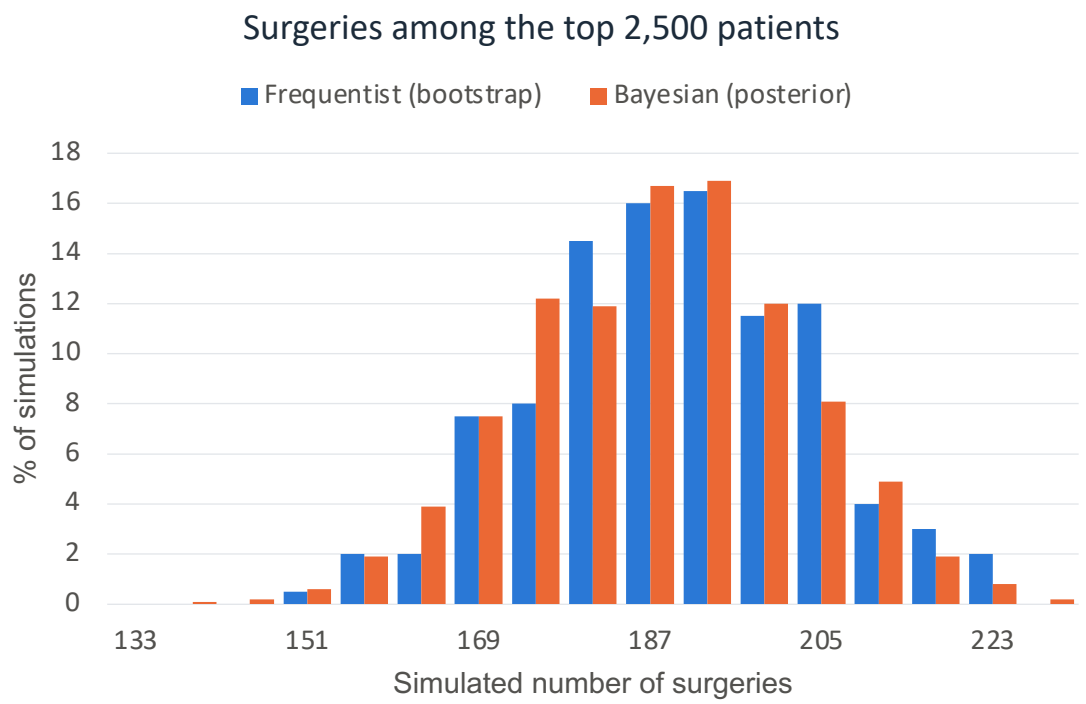
- Read directly off the posterior: prior plus data.
- "Given the data and prior, there is an 89% probability this patient's risk is in this range."
- Median width here: 0.6%. It's narrower because the prior shrinks weak coefficients.

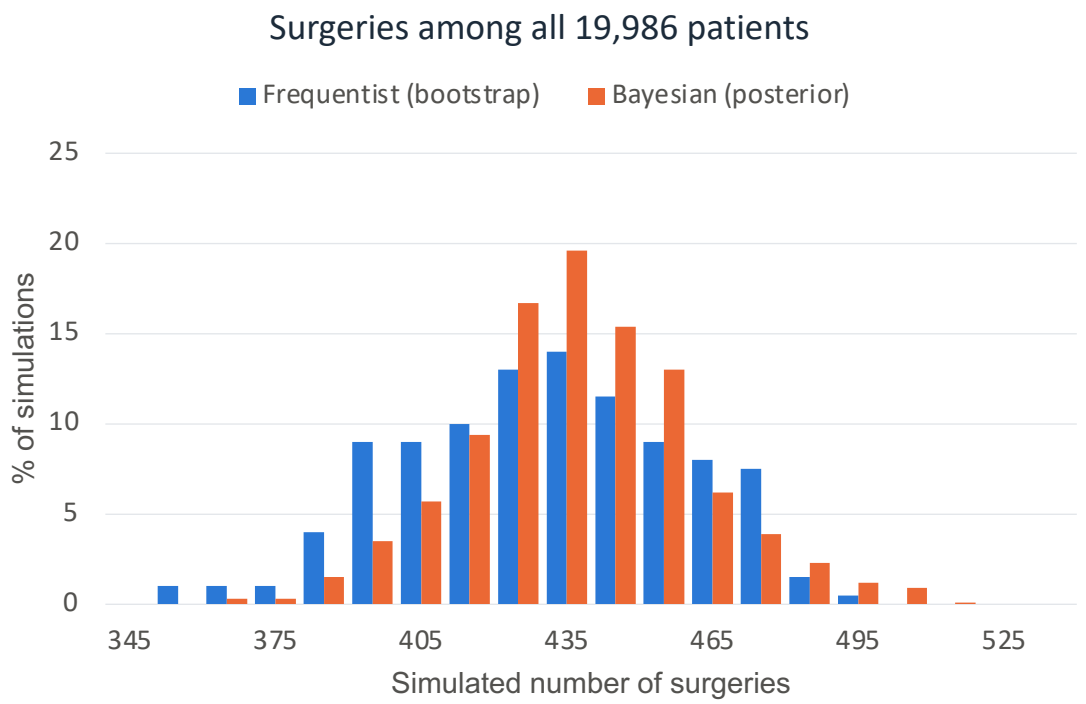
What scikit-learn doesn't give out of the box is the interval: predict_proba returns only a point. The uncertainty still exists; you have to compute it.

Posterior predictive checks

Simulate outcomes from each model and compare with what actually happened in 2025-07

Simulated surgery counts match what happened





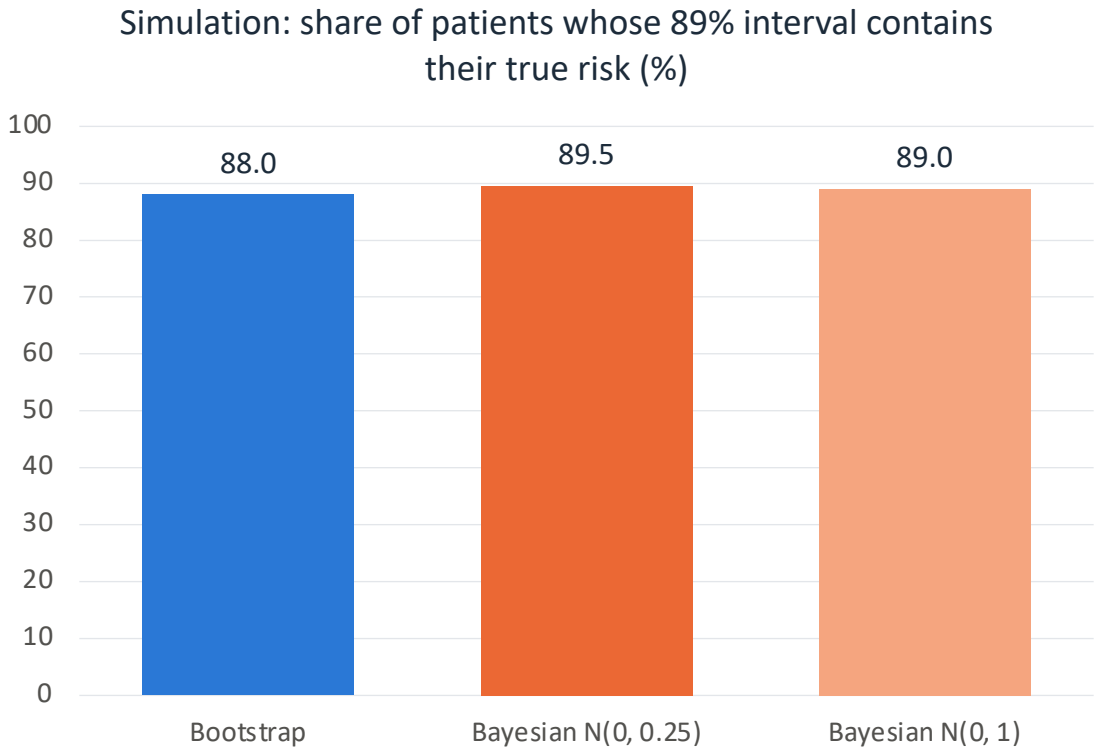
Observed: 179

Bayesian 89% range 164–210, frequentist 166–213. Inside both ranges.

Observed: 431

Bayesian 89% range 399–477, frequentist 387–473. Inside both ranges.

Bayesian intervals are not better calibrated



2025-07 patient groups	Bootstrap	Bayesian	No parameter uncertainty
random, size 1000	100%	100%	100%
random, size 200	96%	96%	96%
risk-sorted, size 1000	79%	79%	79%
risk-sorted, size 200	91%	91%	90%

Real data: share of groups whose observed surgery count falls in the 89% predictive interval.

Simulation (10 replicates): both hit about 89%, and the 1.5-point gap is within replicate noise (each run ranges 72%–95%). Real data: all three columns are identical, because parameter uncertainty is tiny next to outcome noise. The miss in risk-sorted groups is model error that no interval represents.

What only the Bayesian model adds: list certainty

1,799

patients make the top 2,500 in over 90% of posterior draws

1,475

patients make it in 10–90% of draws: effectively interchangeable

4.0%

predicted-risk cut-off for the top 2,500 in 2025-07

How to use it

- Contact the near-certain group every quarter.
- The uncertain group is the natural pool for the pilot's randomized holdout: the data can't separate these patients, so randomizing among them costs almost nothing.
- Report the value of the list as a range, carrying uncertainty in the risk, the enrollment lift and the surgery cost.

Score with frequentist, report uncertainty with Bayesian

- Same accuracy at 1/1000th of the fit time. Simpler to automate.
- Time since first visit (capped at 12m+) fixed the shared 30% over-prediction. The 2026-07 list is low-risk: 919 patients clear the 1.7% break-even.
- Bayesian intervals are no better calibrated. Bayesian earns its place in list certainty, the pilot pool, and a future model across clients.